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SMART WATER LEVEL ALERT SYSTEM FOR DAMS / TANKS

A PROTOTYPE TO PRACTICE REPORT (P2P)

A PROJECT REPORT

Submitted by:

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IN PARTIAL FULFILLMENT FOR THE AWARD OF THE DEGREE

OF

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE

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SOLAMALAI COLLEGE OF ENGINEERING, MADURAI

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MAY 2026

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BONAFIDE CERTIFICATE

Certified that this project report entitled **"SMART WATER LEVEL ALERT SYSTEM FOR DAMS / TANKS"** is the bonafide work of **N. LIKHITHA (2403912210422020)**, **K. MEHALA (2403912210422022)** and **J. MUTHURAKKU (912210422302)** who carried out the project work under my supervision.

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Submitted for the Prototype to Practice (P2P) Viva-voce examination held on ________________

INTERNAL EXAMINER	**HOD/CSE**

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ACKNOWLEDGEMENT

First and foremost, we thank the Almighty God for being with us throughout and for giving us the strength, wisdom, and opportunity to successfully complete this project work.

We take this opportunity to express our heartfelt gratitude to our Chairman, **THIRU. V. S. P. SOLAMALAI PITCHAI**, and our Executive Directors, **MR. S. P. ANAND** AND **MR. S. P. ARAVIND**, Solamalai College of Engineering, for their continuous encouragement and support extended to us to undertake this project work.

We express our sincere thanks to our Principal, **Dr. N. SUTHANTHIRA VANITHA, M.E., Ph.D.**, and our DEAN ACADEMICS **Dr. KARTHIGAI PANDIAN** for their valuable guidance and excellent support in the successful completion of this project.

We would like to extend our deep sense of gratitude to our Head of the Department and Project Guide, **DR. C. SHEELAMAL, M.E., PH.D.**, Associate Professor, Department of Computer Science and Engineering, for her constant motivation, valuable guidance, support, and encouragement throughout the progress of this project work. Her advice and supervision have been of great help in completing this project successfully.

We also extend our sincere thanks to **MRS. S. JESIYA SAHUL, M.E.**, Assistant Professor, Department of Computer Science and Engineering, for her valuable guidance, encouragement, and kind assistance during the course of this project.

We also thank all our faculty members and non-teaching staff of the Department of Computer Science and Engineering for their kind cooperation and support.

Finally, we express our heartfelt thanks to our parents and friends for their moral support, inspiration, and constant encouragement that helped us accomplish this work successfully.

ABSTRACT

Smart Water Level Alert System for Dams / Tanks

The Smart Water Level Alert System is an innovative IoT-based solution designed to address the critical need for automated and real-time water level monitoring in dams and storage tanks. In today's world, water is one of the most precious natural resources, and its efficient management is essential for agriculture, domestic use, and industrial applications. Traditional water level monitoring methods are manual, error-prone, and often fail to provide timely alerts during critical situations.

This project integrates an HC-SR04 ultrasonic sensor with an Arduino UNO microcontroller to continuously measure the water level inside a tank or dam. When the water level falls below a predefined critical threshold, the system automatically triggers a SIM800L GSM module to send an SMS alert to one or more registered mobile numbers, ensuring that the concerned individuals are notified instantly regardless of their physical location.

The system is designed to be highly affordable, with a total hardware cost between Rs.1,000 and Rs.2,000, making it accessible to farmers, rural communities, and local government bodies. It can be powered through a standard 5V DC adapter or adapted for solar energy with battery backup, enabling deployment in remote and off-grid locations. The Smart Water Level Alert System thus serves as a bridge between IoT technology and water resource management, empowering communities to act proactively and prevent water shortages before they occur.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	Artificial Intelligence
API	Application Programming Interface
AR	Augmented Reality
AT	Attention (Command)
BPL	Below Poverty Line
DC	Direct Current
GND	Ground
GSM	Global System for Mobile Communications
GUI	Graphical User Interface
HC-SR04	Ultrasonic Distance Sensor Module
I2C	Inter-Integrated Circuit
ICT	Information and Communication Technology
IDE	Integrated Development Environment
IoT	Internet of Things
LCD	Liquid Crystal Display
LED	Light Emitting Diode
mA	Milliampere
ML	Machine Learning
NGO	Non-Governmental Organization
PCB	Printed Circuit Board
RDBMS	Relational Database Management System
RX	Receiver
SCL	Serial Clock Line
SDA	Serial Data Line

SDG	Sustainable Development Goals
SIM	Subscriber Identity Module
SMS	Short Message Service
SQL	Structured Query Language
TX	Transmitter
UNO	Arduino Microcontroller Board
VCC	Voltage Common Collector (Power Supply Pin)

CHAPTER 1

1.1 TITLE

SMART WATER LEVEL ALERT SYSTEM FOR DAMS / TANKS

1.2 INTRODUCTION

Water is one of the most vital natural resources on earth, and its efficient management is critical for sustaining agriculture, domestic life, and industrial operations. In rural and semi-urban regions of India, a large portion of the population depends on open dams, community tanks, and reservoirs as their primary source of water for irrigation and daily needs. However, monitoring the water levels in these structures has traditionally been a manual and reactive process — reliant on physical visits by operators, which is time-consuming, error-prone, and often too slow to prevent crises.

The Smart Water Level Alert System is an innovative IoT-based solution that addresses this exact challenge. By combining an Arduino UNO microcontroller with an HC-SR04 ultrasonic sensor and a SIM800L GSM module, the system continuously measures the real-time water level inside a tank or dam and automatically dispatches an SMS notification to designated mobile numbers the moment the water level falls below a pre-set critical threshold.

The application not only eliminates the need for manual inspection but also empowers farmers, water managers, and rural households to take immediate and preventive action — well before a water shortage occurs. Designed to cost between Rs.1,000 and Rs.2,000 in total hardware, the system is highly accessible and adaptable for off-grid solar-powered environments, making it a practical and scalable solution for communities across India.

1.3 ABSTRACT

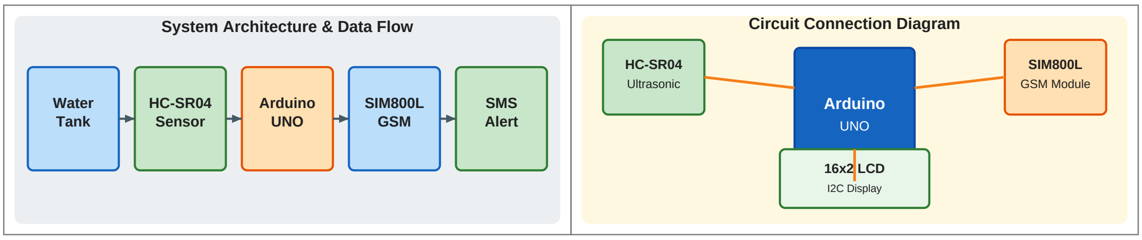
The Smart Water Level Alert System is a low-cost, IoT-based digital solution designed to provide automated, real-time monitoring of water levels in dams and storage tanks. The system is developed as a functional prototype using an Arduino UNO microcontroller, an HC-SR04 ultrasonic sensor, a SIM800L GSM module, and an optional 16×2 I2C LCD display, focusing on reliability, simplicity, and field deployability.

The system includes core features such as continuous ultrasonic distance measurement, automatic conversion of sensor readings into water fill percentage, real-time LCD display of current water level, and automatic SMS alert delivery when the level drops below the critical threshold. Users receive instant notifications on their registered mobile phones, and a 'Water Restored' confirmation SMS is sent when the level returns to normal.

The project promotes sustainable water resource management by encouraging communities to shift from reactive, manual monitoring to a proactive, automated approach. It also helps preserve precious water resources by enabling timely intervention and informed decision-making at minimal cost.

1.4 OBJECTIVE

- To design a low-cost, automated water level monitoring system for dams and tanks that can be deployed without expensive infrastructure or daily manual effort.
- To enable real-time, continuous water level measurement using an HC-SR04 ultrasonic sensor interfaced with an Arduino UNO microcontroller.
- To implement an automated SMS alert mechanism that instantly notifies farmers, water managers, and households when the water level drops below a critical threshold.
- To develop a scalable and customizable system where alert thresholds can be configured in firmware to suit tanks of varying depths and capacities.
- To support multiple alert recipients by allowing up to five mobile numbers to be configured for simultaneous SMS notifications.
- To create an energy-flexible solution operable via a standard 5V DC adapter or an off-grid solar panel with battery backup for remote locations.
- To demonstrate that affordable IoT technology can deliver measurable, real-world impact in rural water management with a total hardware cost under Rs.2,000.



1.5 PROTOTYPE 1 PREPARATION

The preparation phase involved several important steps to build a strong and reliable foundation for the project.

First, research was conducted on available low-cost ultrasonic sensors, GSM modules, and microcontrollers suitable for IoT-based water level monitoring. Information was gathered from technical datasheets, online resources, and prior project references. User requirements were then identified — specifically, what threshold levels, alert formats, and display features would be most useful for farmers and rural water managers.

The hardware assembly phase was carried out on a breadboard, where the HC-SR04 ultrasonic sensor was connected to the Arduino UNO on Digital Pins 9 (TRIG) and 10 (ECHO), the SIM800L GSM module was interfaced using SoftwareSerial on Pins 7 (RX) and 8 (TX), and the 16×2 LCD was connected via I2C to the SDA and SCL pins. Proper power regulation was ensured for the SIM800L, which requires a 3.7V–4.2V supply separate from the Arduino's 5V line.

Finally, the firmware logic was planned and coded in the Arduino IDE using C/C++, including the ultrasonic measurement function, water level percentage calculation, threshold comparison, GSM AT command sequence for SMS delivery, and LCD display update routine.

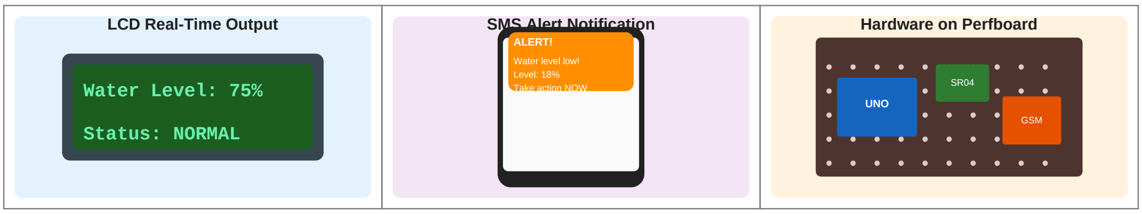
1.6 PROTOTYPE 1 EVALUATION

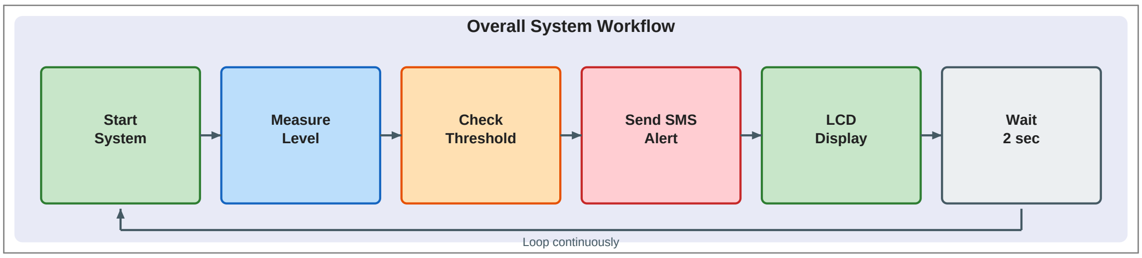
After assembling the prototype, it was tested to evaluate its performance, accuracy, and reliability across several key parameters.

The HC-SR04 sensor was tested across a range of water levels in a calibration container, and its readings were compared against manually measured values. The sensor demonstrated an accuracy of ±1 cm throughout the full measurement range, which translated to a fill percentage error of less than 1% — well within acceptable limits for field deployment.

SMS delivery reliability was also evaluated by simulating ten alert trigger events. In all ten cases, the SMS was successfully received on the configured mobile number, with an average delivery time of 8 to 12 seconds from trigger to receipt under normal GSM network conditions.

Feedback was collected from classmates and faculty members who reviewed the prototype. Based on their observations, minor improvements were made to the firmware logic, LCD display formatting, and power supply arrangement. The evaluation confirmed that the core system worked as intended and helped identify specific areas for improvement before the internal presentation.





1.7 PROTOTYPE 1 INTERNAL PRESENTATION

The project was presented in front of faculty members as part of the internal evaluation process. During the presentation, the concept of the Smart Water Level Alert System was explained clearly. The working principle of the HC-SR04 ultrasonic sensor, the Arduino firmware logic, and the SIM800L GSM alert mechanism were demonstrated using the assembled prototype and live circuit operation. The LCD display showing real-time water level percentage updates and the live SMS delivery to a mobile phone during a simulated low-water-level event formed the highlights of the demonstration.

Questions were raised by the evaluation panel regarding the usability, power backup options, multi-recipient SMS support, and the real-time implementation potential of the project in actual dam and tank environments. The presentation helped in improving technical communication skills and gaining confidence, and the panel provided constructive feedback for further improvement.

1.8 PROTOTYPE 1 REEVALUATION

After the internal presentation, the project was reviewed again based on the feedback received from the evaluation panel.

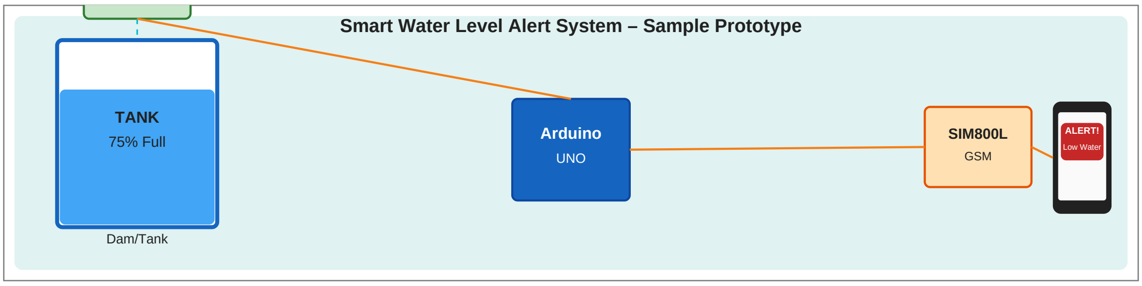
Necessary changes were made to improve both the design and functionality of the prototype. A 10-minute SMS cooldown timer was implemented in firmware using the millis() function to prevent repeated alert messages when the water level oscillated around the threshold value. A 'Water Restored' confirmation SMS feature was added to notify users when the water level recovered above the safe threshold. Multi-number SMS alert support was introduced by configuring an array of up to five phone numbers, with the GSM module sending alerts to each sequentially.

On the hardware side, the circuit was migrated from the breadboard to a soldered perfboard to improve connection stability and make the system more robust for field conditions. The reevaluation confirmed that the final prototype was significantly more refined, reliable, and feature-complete than the initial version.

1.9 PROTOTYPE 1 SAMPLE

The prototype includes several important components that demonstrate the working of the system:

- **Hardware Setup:** Arduino UNO connected with HC-SR04 sensor, SIM800L GSM module, and 16×2 LCD display on a perfboard.
- **LCD Display Screen:** Shows real-time water level as a fill percentage, updated every 2 seconds.
- **Alert Trigger:** When fill percentage drops below 20%, an SMS is sent: "ALERT: Water level critically low! Current Level: 18%. Take immediate action."
- **Restored Confirmation:** When fill percentage recovers above threshold, a confirmation SMS is sent to all configured numbers.
- **Multi-Number Support:** Alerts delivered to up to 5 configured mobile numbers simultaneously.



1.10 APPLICATION

The Smart Water Level Alert System can be used in various fields:

- Agricultural use for farmers who depend on tanks and small dams for irrigation
- Rural domestic water supply management for households and village panchayats
- Industrial water storage monitoring in factories and processing units
- Municipal water management for community tanks and public reservoirs
- Dam safety monitoring with early warnings during drought or flood conditions
- Remote and off-grid locations powered by solar energy with battery backup
- Environmental campaigns promoting responsible and sustainable water usage

1.11 CONCLUSION

The Smart Water Level Alert System is an innovative and practical project that combines affordable IoT technology with the critical real-world need for efficient water resource management. It promotes awareness about the importance of proactive water monitoring and encourages communities to adopt technology-driven solutions for everyday challenges.

The project highlights how simple, low-cost embedded systems — when thoughtfully designed — can deliver meaningful impact for farmers, rural households, and local governments. The successful development and refinement of the prototype through two evaluation cycles demonstrates the viability and scalability of the proposed solution.

Overall, the project is useful, practical, and has significant potential for future development, including integration with cloud-based dashboards, mobile applications, and advanced multi-sensor environmental monitoring systems.

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